WATER RESOURCES, WELLS & PUMPS · NOTE WR03

Irrigation Terms and Project Classes — GCA, CCA, Potential, and Major, Medium and Minor Schemes

Course topics: IDE5.5 L2, IDE8.16 L2, IDE8.16 L4, IDE-503 T2

Command-area terms

Term	Meaning
Gross command area (GCA)	Total area a channel or project can irrigate — including cultivable land, roads, culverts, villages, unculturable areas and anything else not available for cultivation
Culturable command area (CCA)	The culturable (cultivable) part of the GCA
Intensity of irrigation	The area actually irrigated in a year from an outlet, as a share of the CCA. In most canals of northern India it falls short of the CCA
Irrigation potential	Gross area that can be irrigated with the water available under the planned cropping pattern
Utilisation	Gross area actually irrigated each year

$$\text{CCA} = \text{GCA} - \text{unculturable area in the GCA}$$

> Bhakra Canal Circle: sanctioned intensity of irrigation 62% — kharif (monsoon crop) 27.6% and rabi (winter crop) 34.4%.

How potential and utilisation are counted

- Potential is "created" once the project is commissioned and water courses up to outlet level are ready.
- A single watering in kharif plus a single watering in rabi both count, so potential created can be twice the CCA — even though one watering may not meet the crop's need.
- Full utilisation needs the project commissioned and field channels built beyond the outlet. In years of good rain a crop may need no irrigation, so no utilisation is recorded — and departments do not record such cases.
- **Limits of the statistics:** they cover only potential created and utilised; the number of irrigations, adequacy of supply, rise in productivity and efficiency are generally not recorded, so project performance cannot be properly evaluated.

Accelerated Irrigation Benefit Programme (AIBP): launched by the Government of India in 1996–97 to speed up ongoing irrigation projects with financial help to the States.

Storage potential

- Potential storage of surface flow in India's river basins: 378 cubic km, against total water availability of 1869 cubic km.

- The Ganga basin has the most storage; Krishna, Godavari, Narmada and Mahanadi also have large capacities. Including projects under consideration, the Brahmaputra and Barak basin ranks second after the Ganga.
- As a share of average annual flow, the Tapi basin leads; the west-flowing rivers (including the Narmada) exceed 50%; the Ganga basin is only about 13%.

Classification of irrigation projects

India classified projects by CCA before 1951, by cost from 1951 to 1978, and by CCA again since 1978.

Class	CCA	Nature
Major	≥ 10,000 ha	Surface water projects with large storage and/or diversion works
Medium	2000 ha to 10,000 ha	Usually surface water; some are large lift schemes
Minor	up to 2000 ha	(a) surface-water minor schemes; (b) groundwater minor schemes

Water resources projects

- Broadly, irrigation projects and hydroelectric projects, planned for one or more of: irrigation, hydropower, drinking water, industrial water, flood control, navigation.
- **Multipurpose projects** serve more than one purpose — most combine irrigation and hydropower.
- Many projects approved independently were later run together after interstate agreements and new upstream and downstream works; they are now described as irrigation, hydropower, multipurpose or complex projects.

Minor irrigation schemes

- Comprise surface-water schemes (flow or lift) and groundwater schemes. Ultimate irrigation potential of surface-water minor schemes: 17.4 million ha.
- They are useful even inside major and medium commands because the farmer controls the water.

Flow scheme	Features
Storage schemes	Store water in tanks, ponds and reservoirs; suit undulating topography and rocky strata. Tanks commanding up to 20 ha belong to individuals, farmer groups or Panchayats (common in Tamil Nadu, Karnataka, Andhra Pradesh, Maharashtra, Odisha, MP and Bihar). Ponds command up to 2 ha (common in the alluvial plains of UP, Bihar and West Bengal)
Diversion schemes	Divert stream water by gravity without storage, using a weir or bund across the stream; more economical, but depend on stream flow during the irrigation season. A weir is called an **anicut** in South India, **bandhara** in Maharashtra and Gujarat and **thingel** in the Assam region. Often a flow-cum-lift system works best

Groundwater irrigation

- Wells in India up to March 1994: 10.22 million dugwells, 5.04 million shallow tubewells and 0.07 million deep (public) tubewells.
- Tamil Nadu had 15% of the dugwells; Maharashtra, MP and Andhra Pradesh 13% each; Uttar Pradesh 50% of the shallow tubewells. Public tubewells lie mainly in the Indo-Gangetic plains.

Update — 6th Minor Irrigation Census (reference year 2017–18, released 2023): 23.14 million minor irrigation schemes — 21.93 million groundwater (about 8.28 million dugwells, 5.59 million shallow, 4.32 million medium and 3.75 million deep tubewells) and 1.21 million surface-water schemes. Groundwater schemes are about 94.8% of the total.

Micro irrigation applies small quantities of water frequently, on or below the soil surface, as drops, tiny streams or miniature spray, through emitters placed along a delivery line.

Sources: ICAR eCourse 'Minor Irrigation and Command Area Development' — Dr K. Shanmuga Sundaram (TNAU), reviewed by Dr P. Balakrishnan (UAS Karnataka). ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). 6th Minor Irrigation Census (Ministry of Jal Shakti, 2023), Rajya Sabha reply. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. CCA equals

(a) GCA + unculturable area
(b) GCA - unculturable area
(c) GCA × intensity
(d) irrigated area - GCA

Q2. A project with CCA of 5000 ha is classed as

(a) major
(b) medium
(c) minor
(d) micro

Q3. Major irrigation projects have a CCA of at least

(a) 2000 ha
(b) 5000 ha
(c) 10,000 ha
(d) 50,000 ha

Q4. Irrigation potential created can be up to ___ the CCA

(a) half
(b) equal to
(c) twice
(d) four times

Q5. AIBP was launched in

(a) 1951–52
(b) 1978–79
(c) 1996–97
(d) 2015–16

Q6. A diversion weir is called "bandhara" in

(a) Tamil Nadu
(b) Maharashtra and Gujarat
(c) Assam
(d) Punjab

Answers

Q1 — (b) GCA - unculturable area
Q2 — (b) medium — *2000–10,000 ha = medium.*
Q3 — (c) 10,000 ha
Q4 — (c) twice — *One kharif and one rabi watering both count.*
Q5 — (c) 1996–97
Q6 — (b) Maharashtra and Gujarat